

Analysis of Residual Block from Problem 2:

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(a) Projection Shortcut Analysis:

The skip connection needs to match dimensions from input to output for addition.

Input: 128 feature maps

Output: 256 feature maps

First conv uses stride=2, which reduces spatial dimensions by half.

Two issues that prevent direct addition:

1. Different number of feature maps (128 vs 256)
2. Different spatial dimensions due to stride=2

Required projection shortcut:

A 1x1 convolution with stride=2 that projects 128 feature maps to 256 feature maps

while also reducing spatial dimensions by half to match the main path.

(b) Parameter Count Calculation:

First conv layer (3x3, stride=2, 128→256):

Filter size: 3x3x128

Number of filters: 256

Parameters: $3 \times 3 \times 128 \times 256 = 294,912$

Second conv layer (3x3, stride=1, 256→256):

Filter size: 3x3x256

Number of filters: 256

Parameters: $3 \times 3 \times 256 \times 256 = 589,824$

Projection shortcut (1x1, stride=2, 128→256):

Filter size: 1x1x128

Number of filters: 256

Parameters: $1 \times 1 \times 128 \times 256 = 32,768$

Total trainable parameters: $294,912 + 589,824 + 32,768 = 917,504$